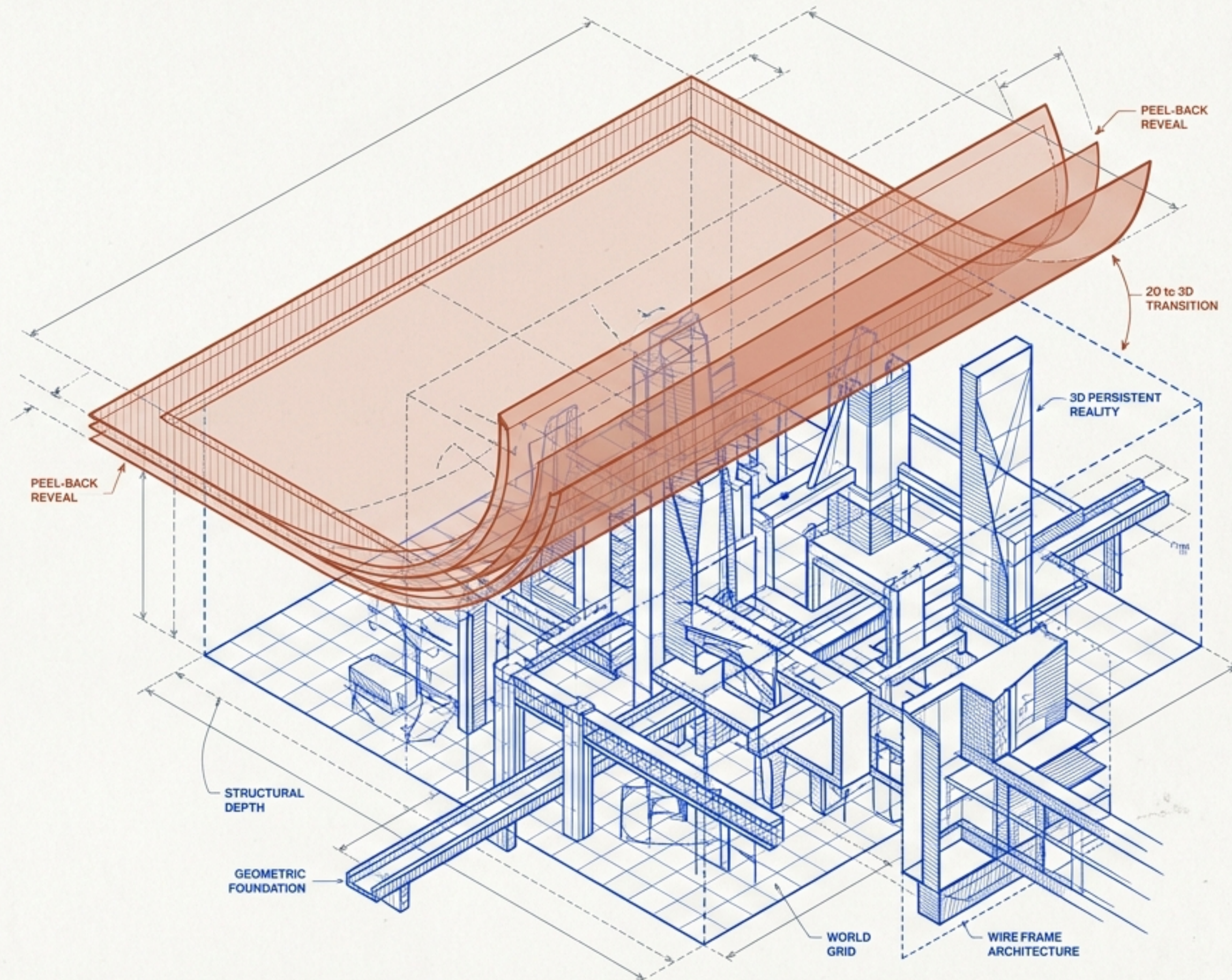


Cinematic Reconstruction

Beyond the Frame:
A Persistent World
Framework for Video
Generation,
Compression,
Compression, and
Diagnosis.

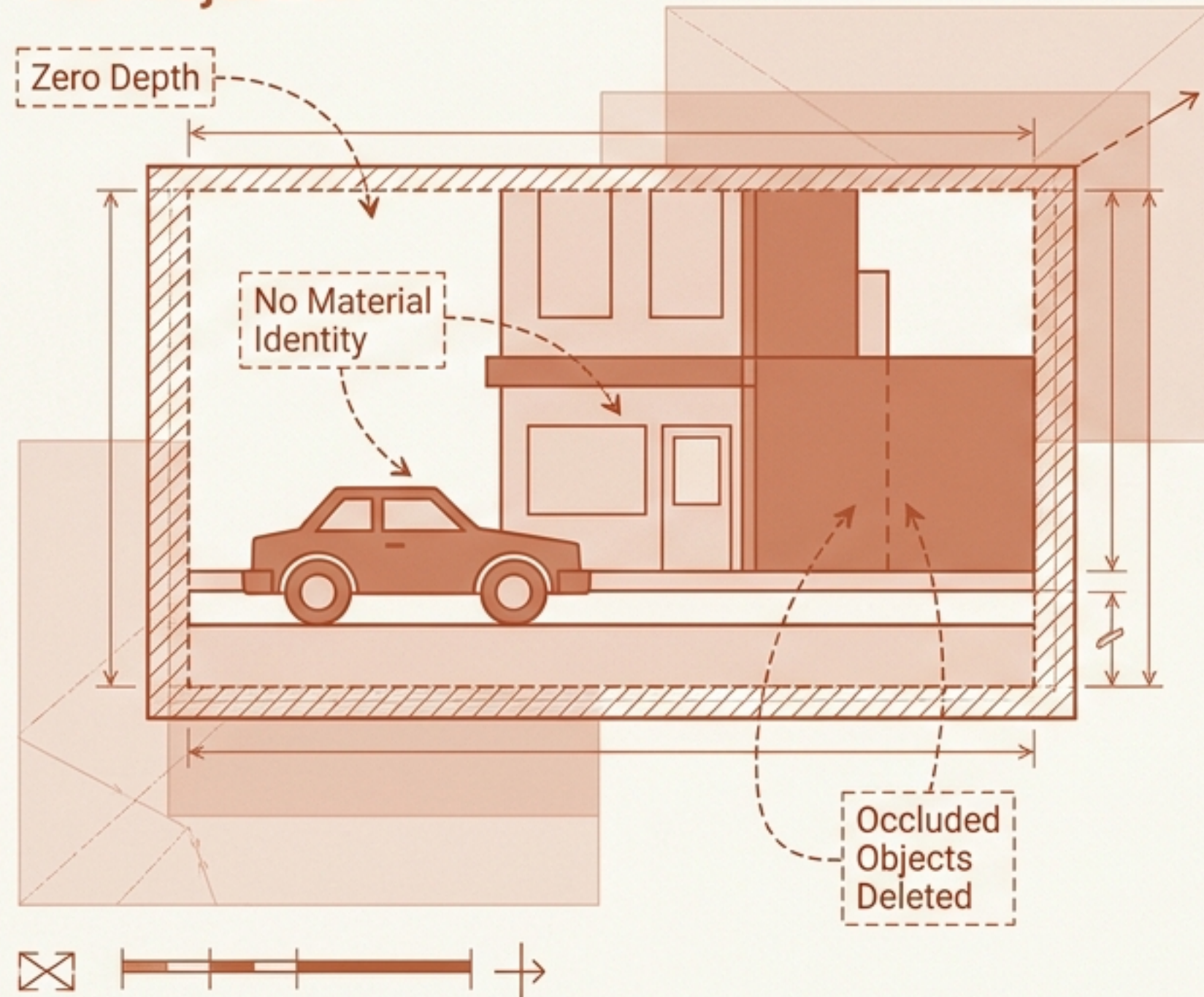
Escaping the "Premature Commitment"
of modern generative AI.



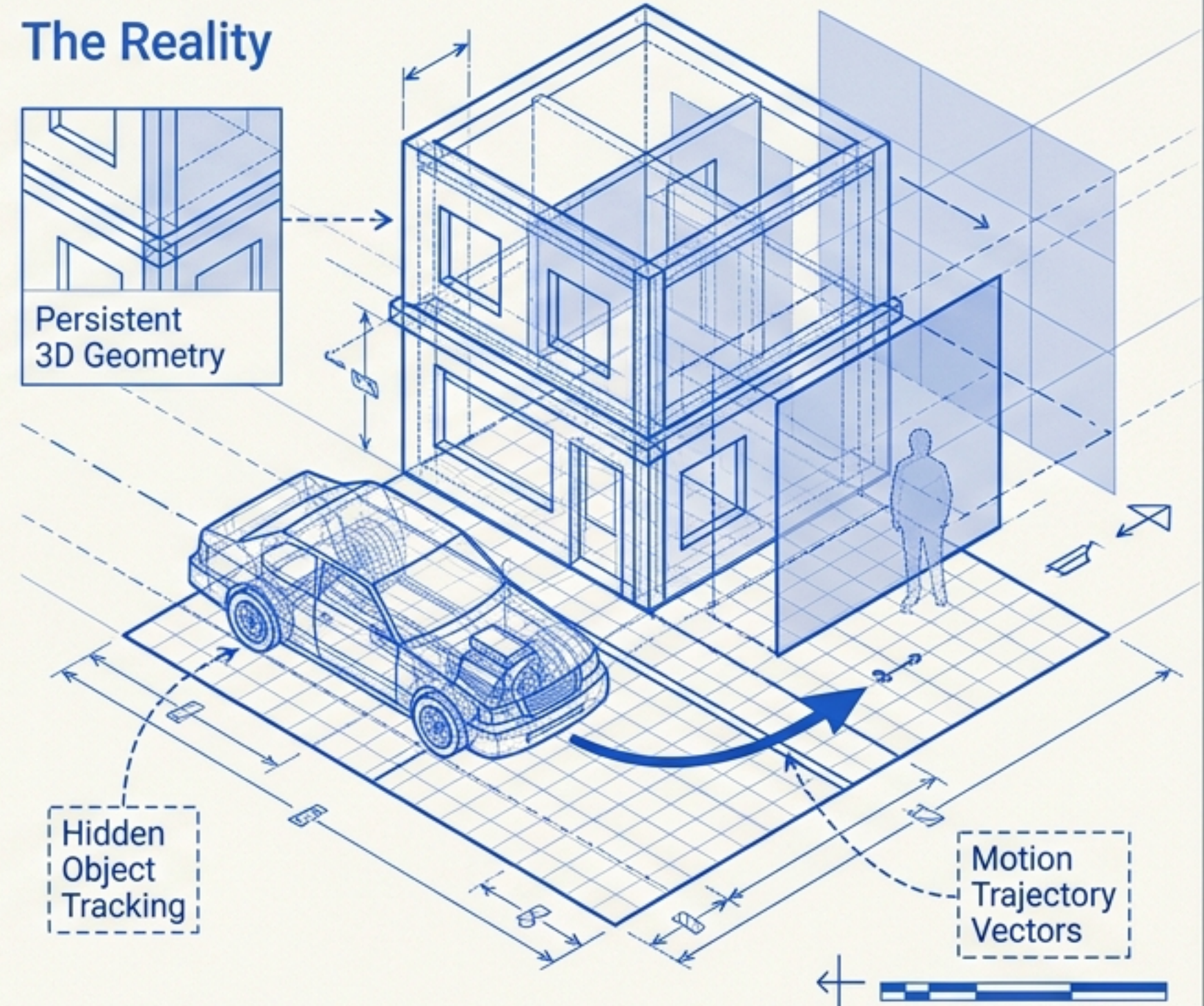
The Contingency of Current Conventions

Modern generative AI and video codecs treat the 2D frame as the fundamental unit of video. This is an architectural trap. A frame is merely a projection—an observation map that discards vital structural data. We have built an industry optimizing the illusion rather than modeling the underlying reality.

The Projection

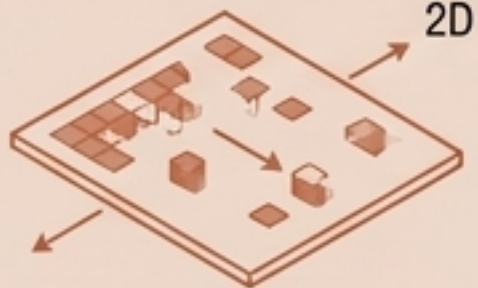
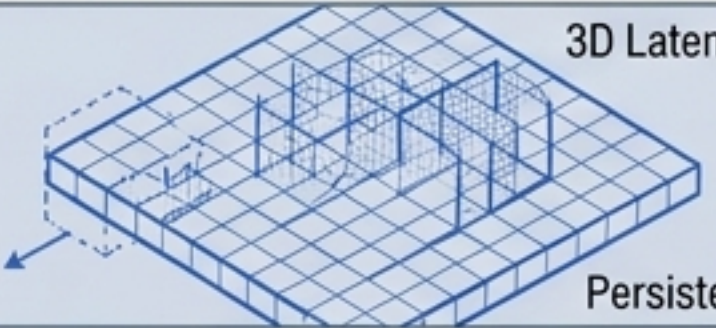
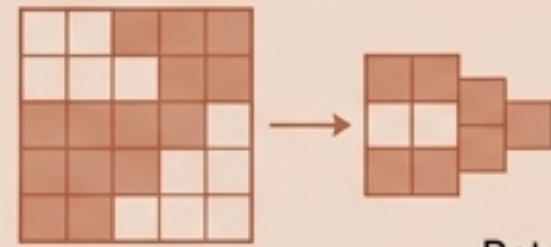
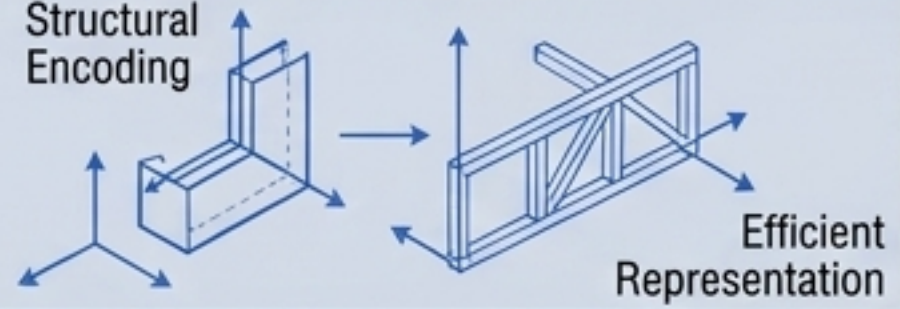
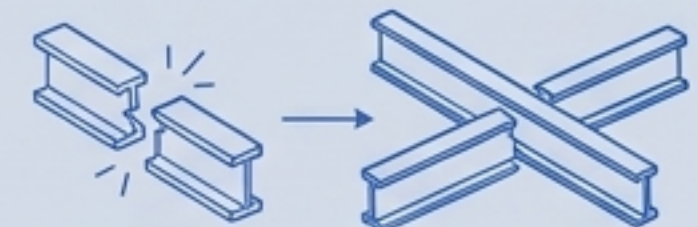
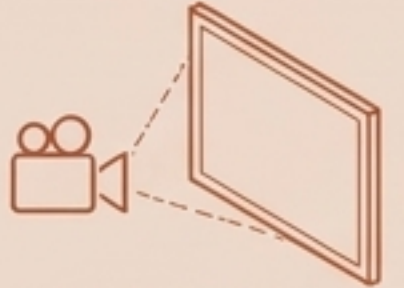
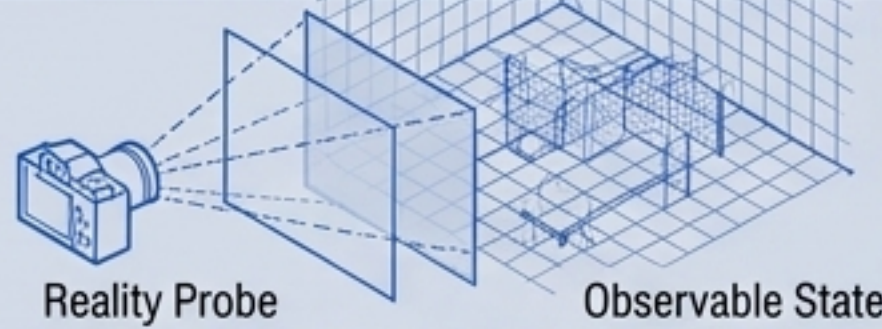


The Reality



The Architectural Paradigm Shift

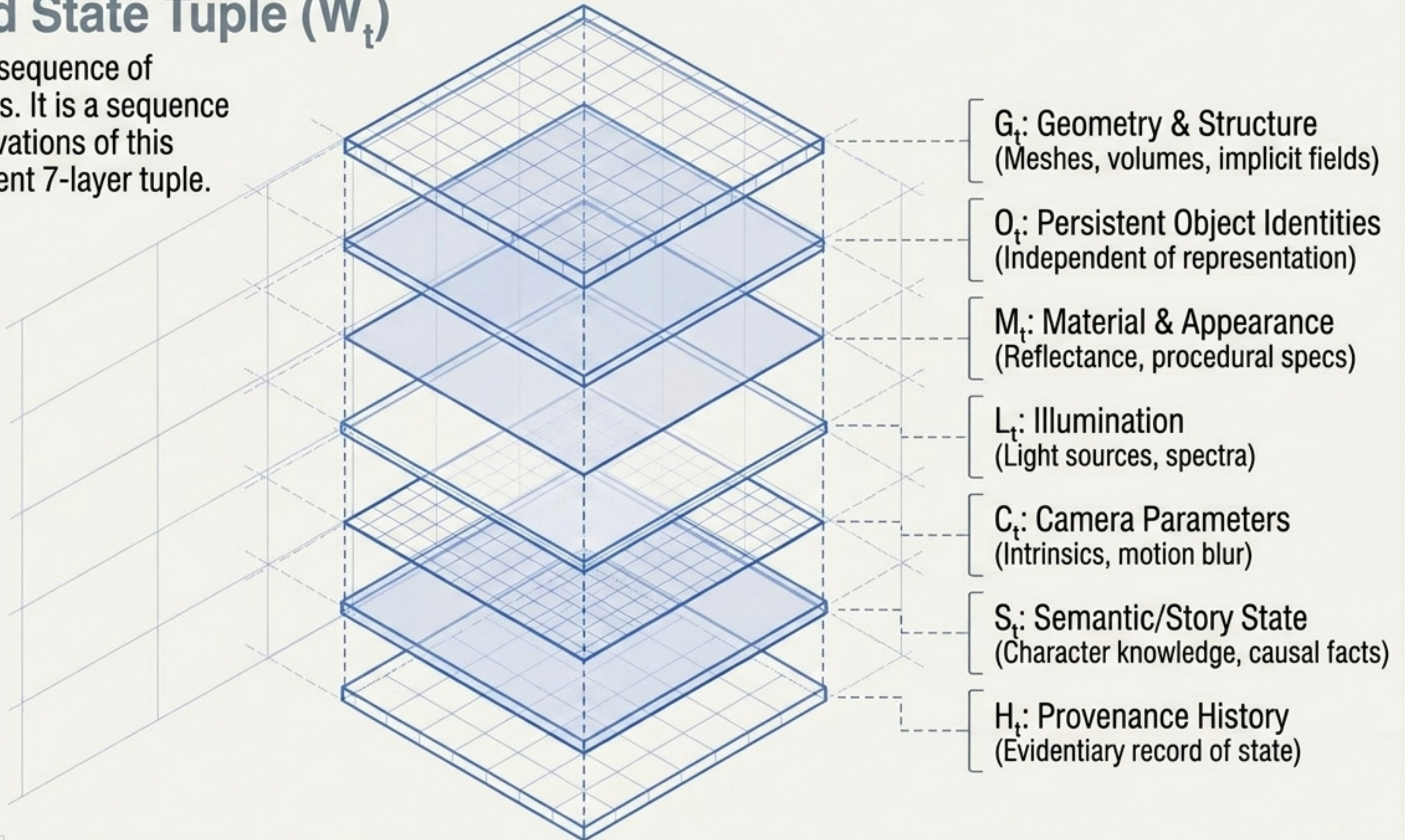


	Current: Frame-Centric	Proposed: World-Centric
Generation	Prompt → Sequence of pixels  2D Surface Output	Operator → Admissible World Continuation  3D Latent Reality Persistent State
Compression	Pixel-block reduction  Data Loss	World-state parameter description  Structural Encoding Efficient Representation
Glitches	Aesthetic artifacts to be smoothed  Visual Noise Illusion Break	Structural evidence of model failure  Model Error Inconsistent Physics
Camera	The creator of the scene  Scene Fabrication	An epistemic observation operator of a latent reality  Reality Probe Observable State

Key Takeaway: Shifting the fundamental unit of computation from the frame to the world solves generative problems that are mathematically unsolvable in 2D space.

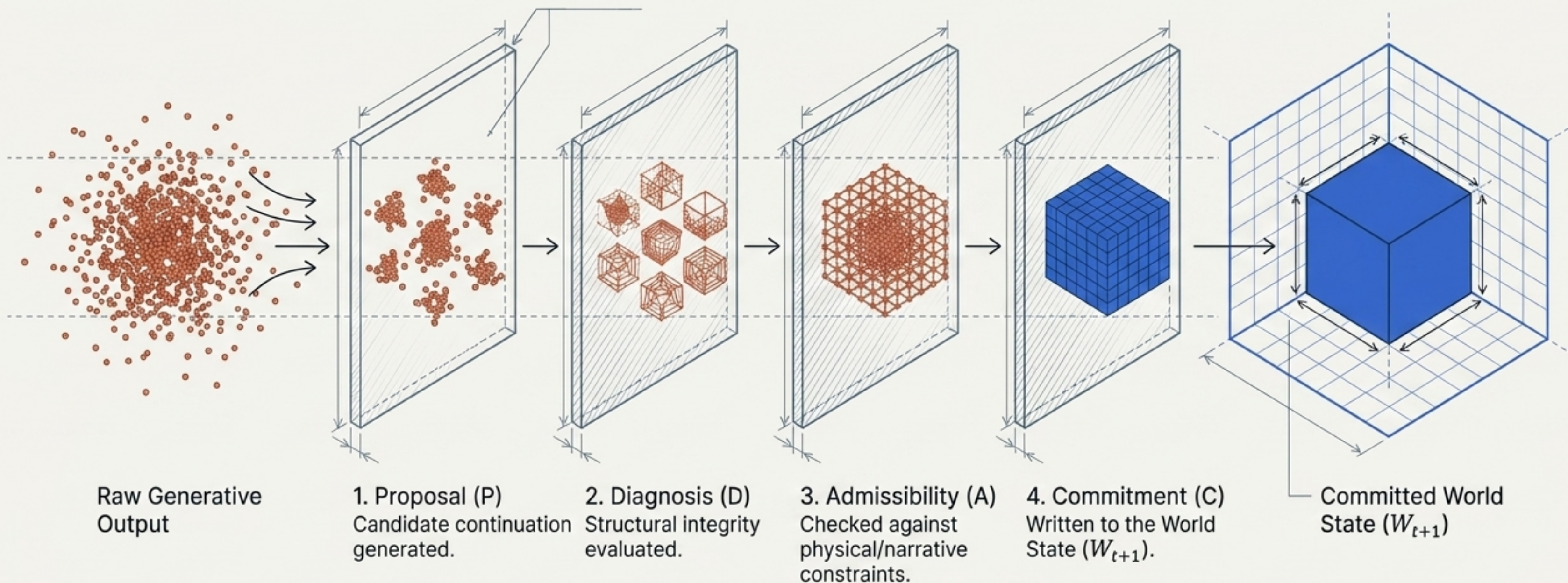
The World State Tuple (W_t)

A movie is not a sequence of generated images. It is a sequence of camera observations of this evolving, persistent 7-layer tuple.



The Probate Mechanism

Generative AI amplifies downstream selection. Unchecked, it normalizes pathology. The architecture interposes a strict 'Probate' mechanism: generated candidates must pass structural, physical, and causal admissibility checks before they are granted authority to update the world.

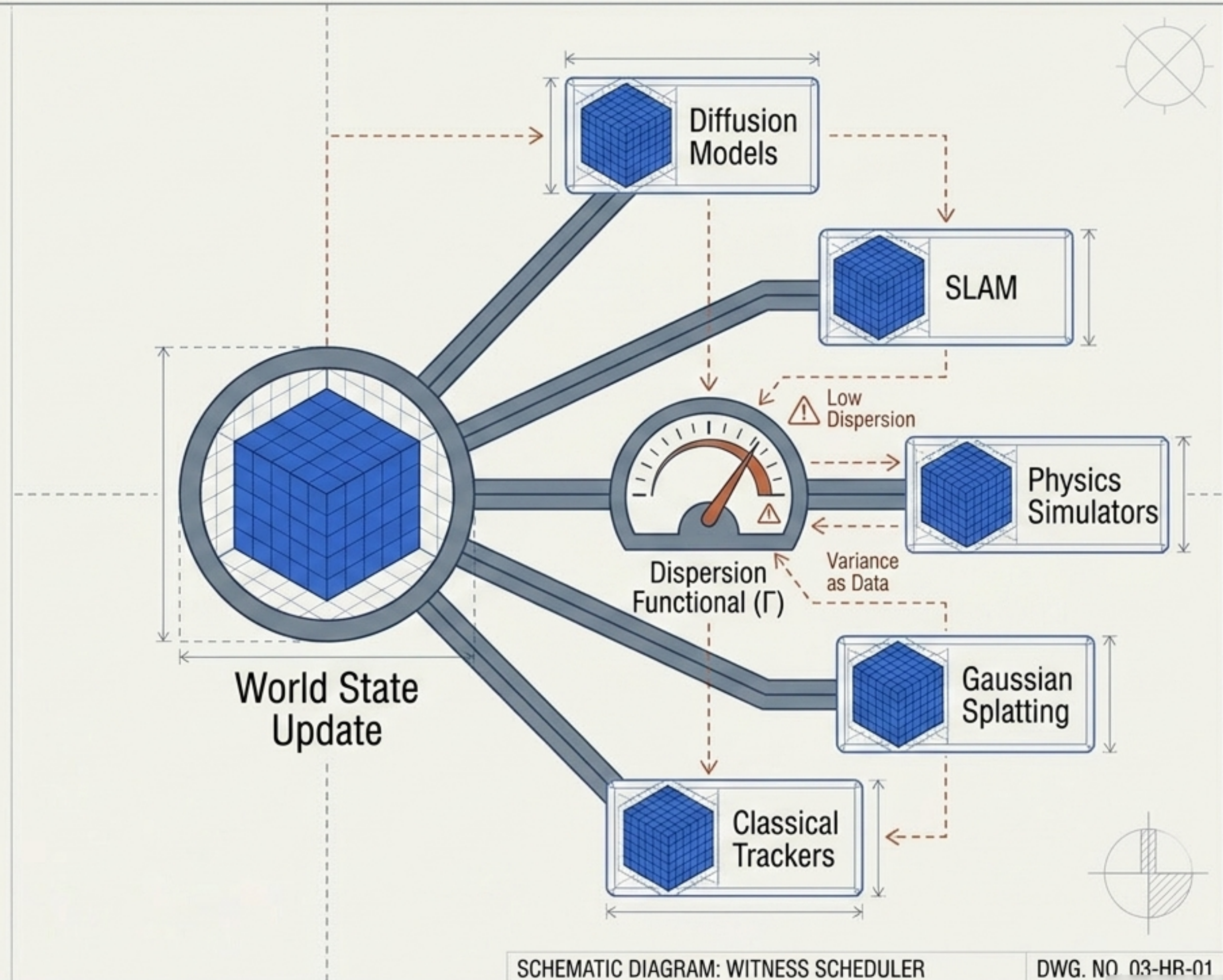


The Heterogeneous Repertoire

No privileged algorithm. Different problems demand different tools.

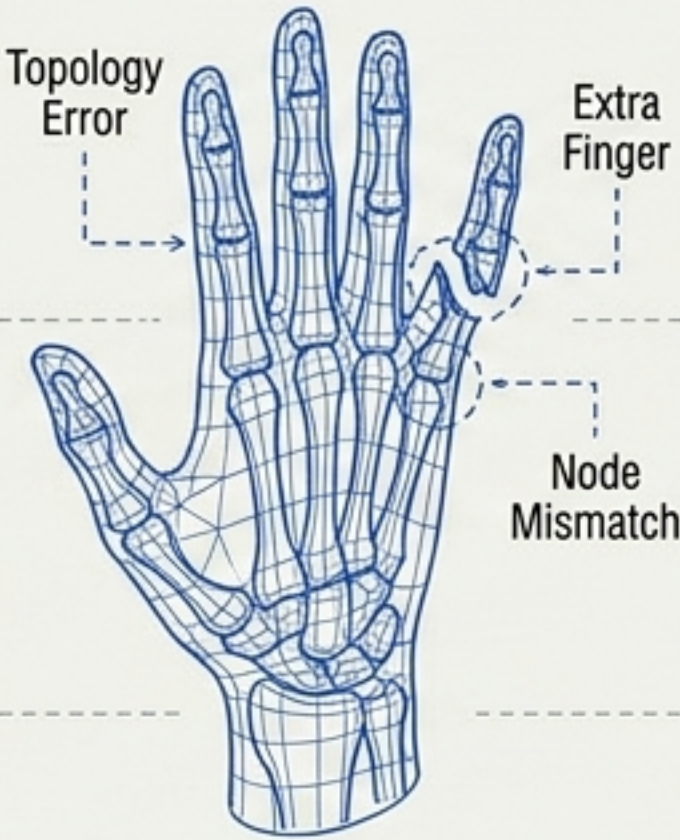
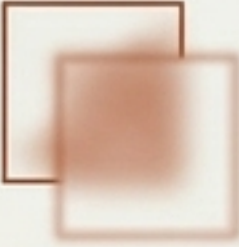
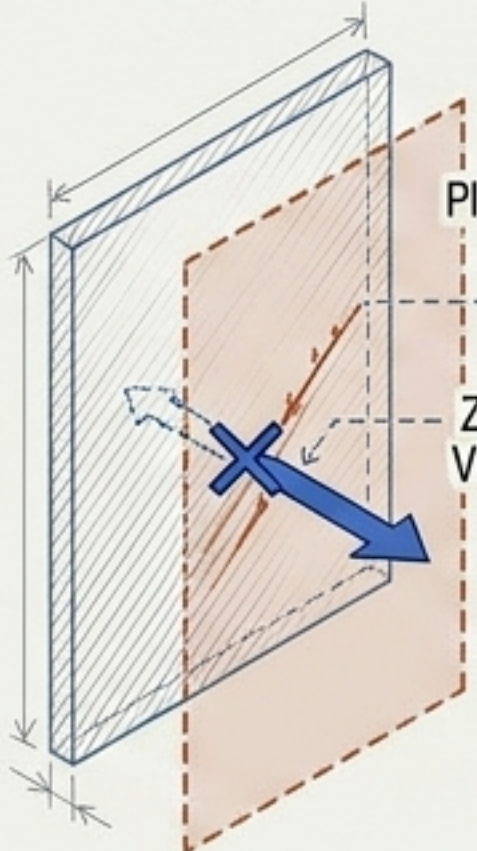
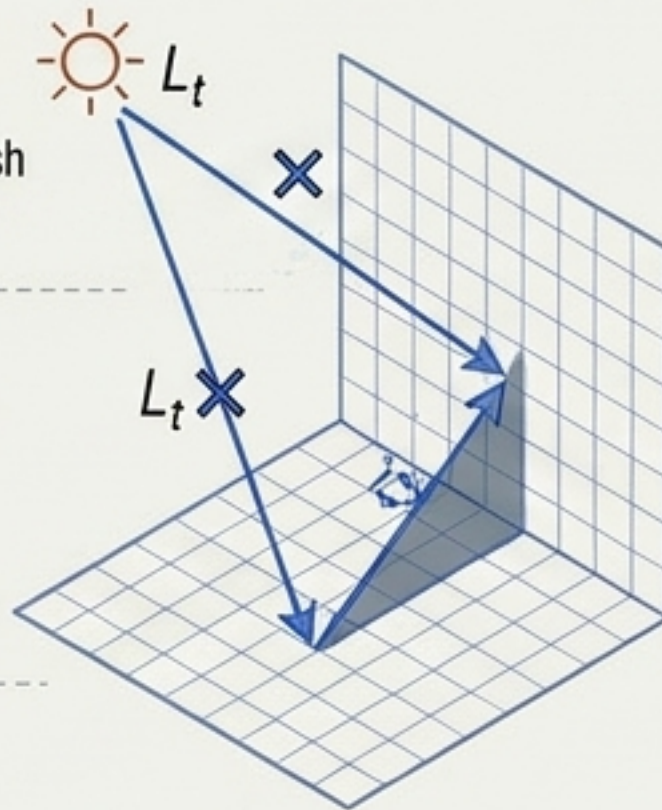
Rather than one massive neural network predicting everything, a Witness Scheduler summons the right algorithm.

Crucially, when independent algorithms disagree on a pixel, this dispersion is treated as vital diagnostic data, not just variance to average away.



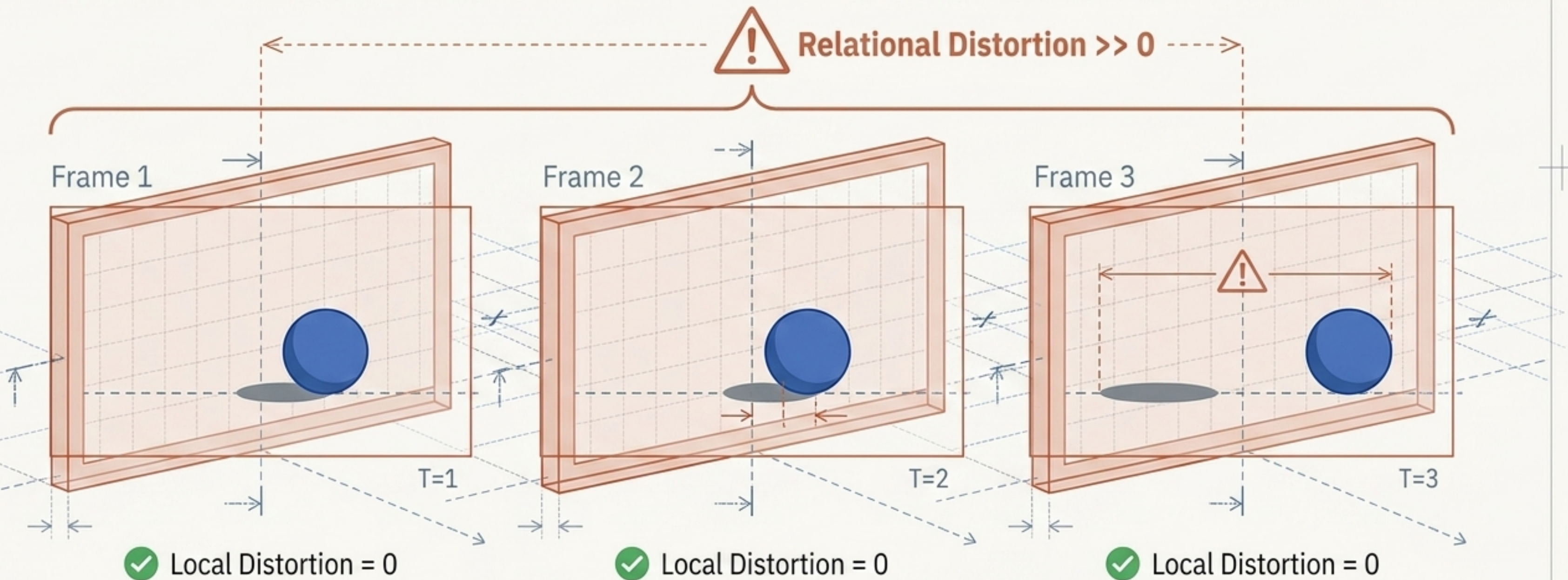
Redefining the Glitch

Glitches are not merely “ugly pixels.” They are precise violations of preserved structural invariants. Defined structurally, they act as forensic evidence, telling the system exactly which representation or algorithm failed in the pipeline.

Symptom	Aesthetic View	Structural Predicate Violated
 Extra Fingers	Bad pixels	 Topology Error Extra Finger Node Mismatch
 Texture Leakage	Blurry edges	 Depth Plane Clash Z-Order Violation
 Impossible Shadows	Ugly lighting	 Illumination (L_t) Inconsistency

Process Grammars & Relational Anomalies

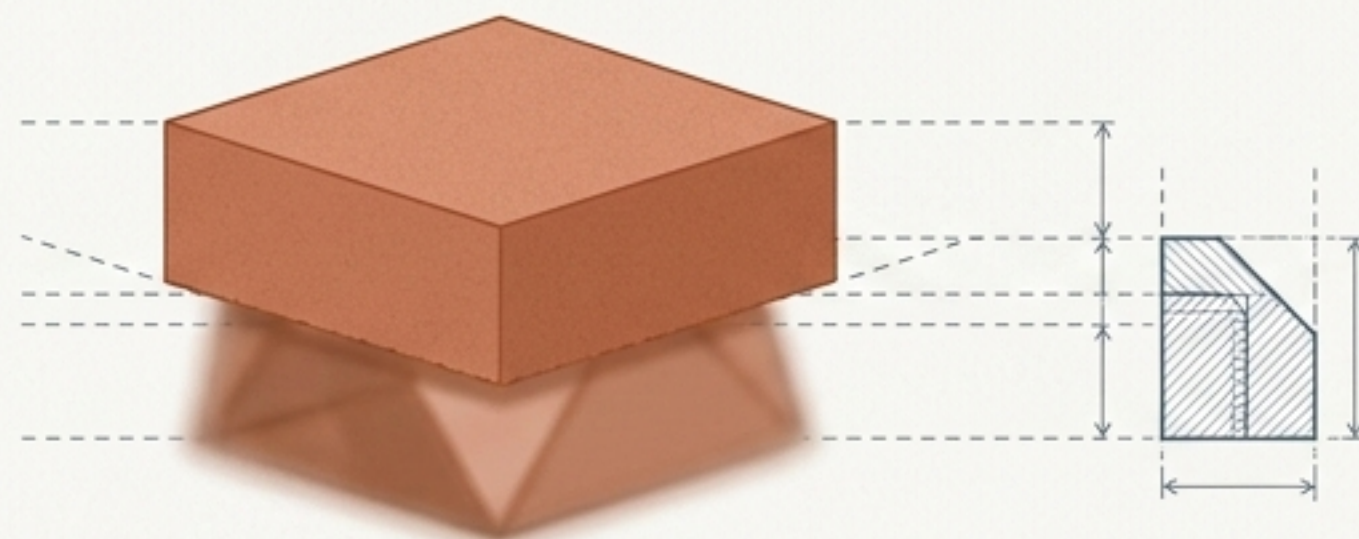
Many cinematic errors are locally plausible but relationally inconsistent. By maintaining a provenance history, we build "Process Grammars" to detect anomalies over time. If a policy consistently requires downstream repairs, it signals a fundamental mismatch, preventing the dangerous normalization of pathology.



Structure-Aware Compression

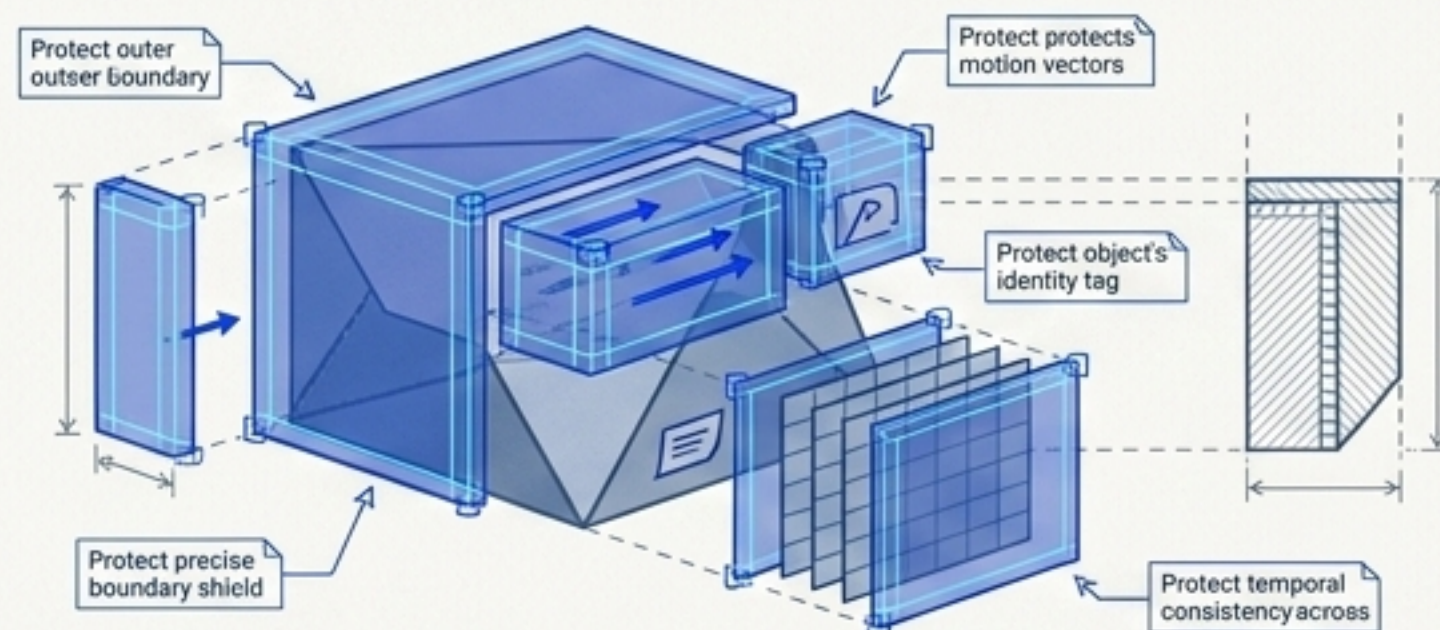
Current codecs degrade uniformly because they only understand pixels. Structure-Aware compression leverages the 3D world-state. Under extreme constraints, structural codecs fail gracefully by preserving object boundaries, identities, and temporal coherence, rather than blurring across realities.

Conventional
Compression



$$R + \lambda D_{\text{pixel}}$$

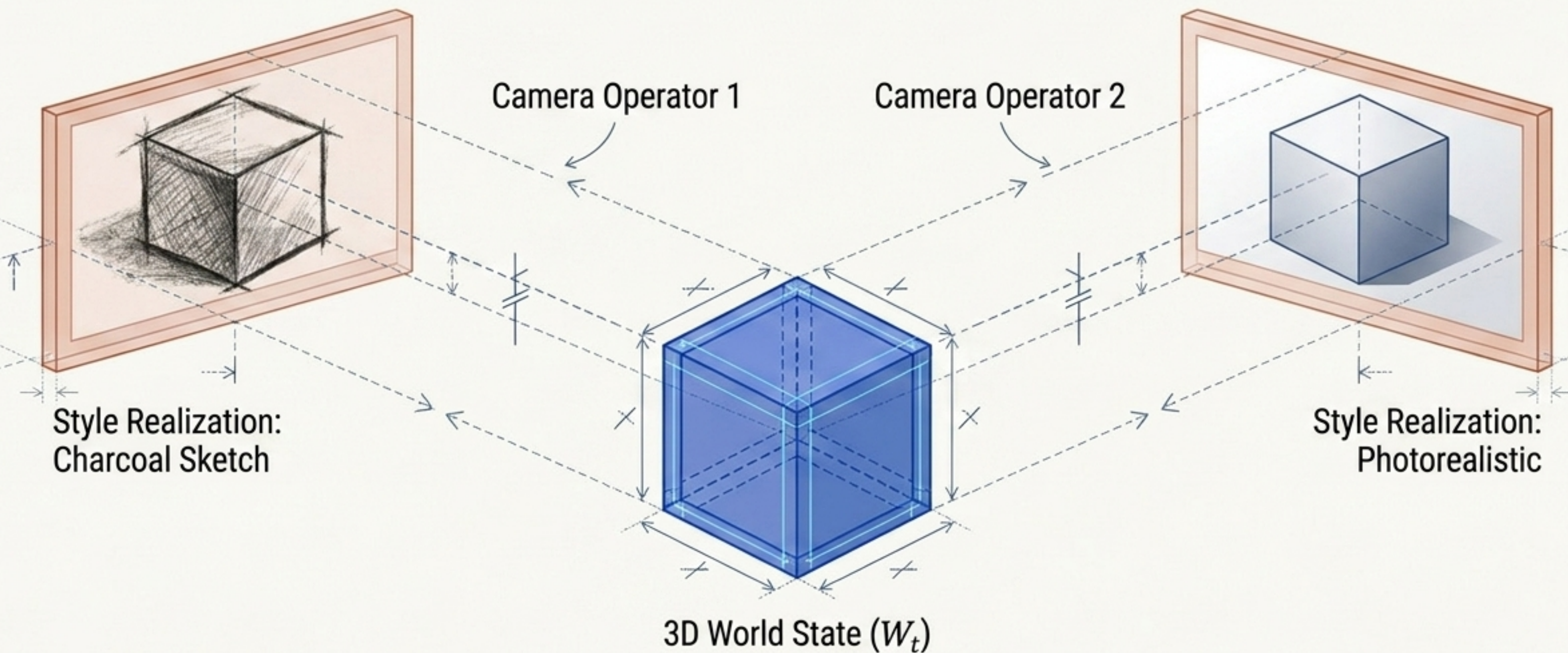
Proposed
Structure-Aware
Compression



$$R + \underbrace{\lambda_1 D_{\text{app}}}_{\text{Protect precise modular shield}} + \underbrace{\lambda_2 D_{\text{bound}}}_{\text{Protect precise modular shield}} + \underbrace{\lambda_3 D_{\text{motion}}}_{\text{Protect temporal consistency shield}} \\ + \underbrace{\lambda_4 D_{\text{ident}}}_{\text{Protect object's identity tag}} + \underbrace{\lambda_5 D_{\text{depth}}}_{\text{Protect temporal consistency across}} + \underbrace{\lambda_6 D_{\text{temporal}}}_{\text{Protect temporal consistency across}}$$

Separating Identity from Realization

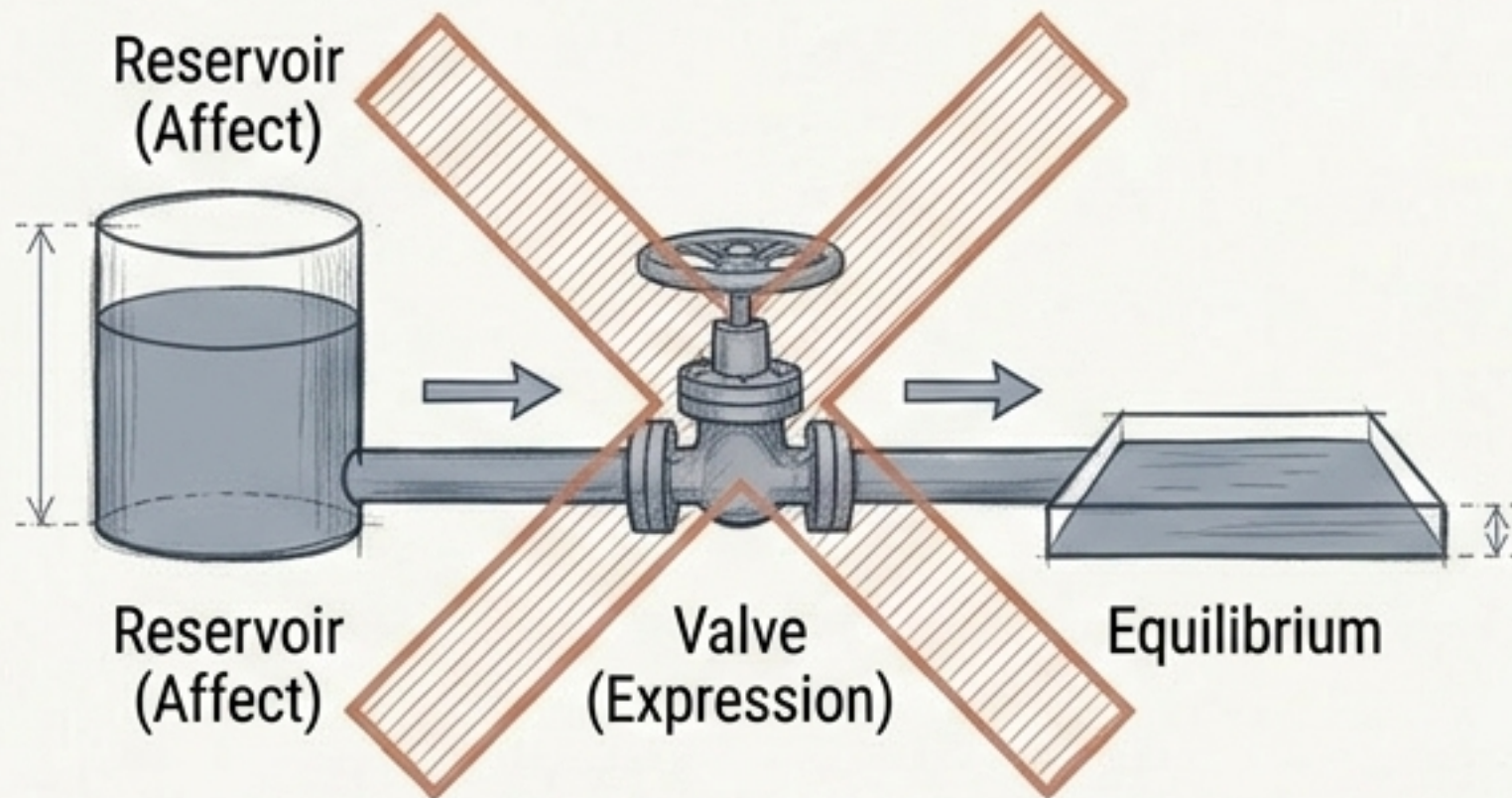
Style is a realization operator that changes visual output without changing the world. The camera is an epistemic operator that changes what the viewer can infer without changing latent reality. Photorealistic or pixel-art, the underlying dog, table, and physics remain identical. Changing style must **never** change the world.



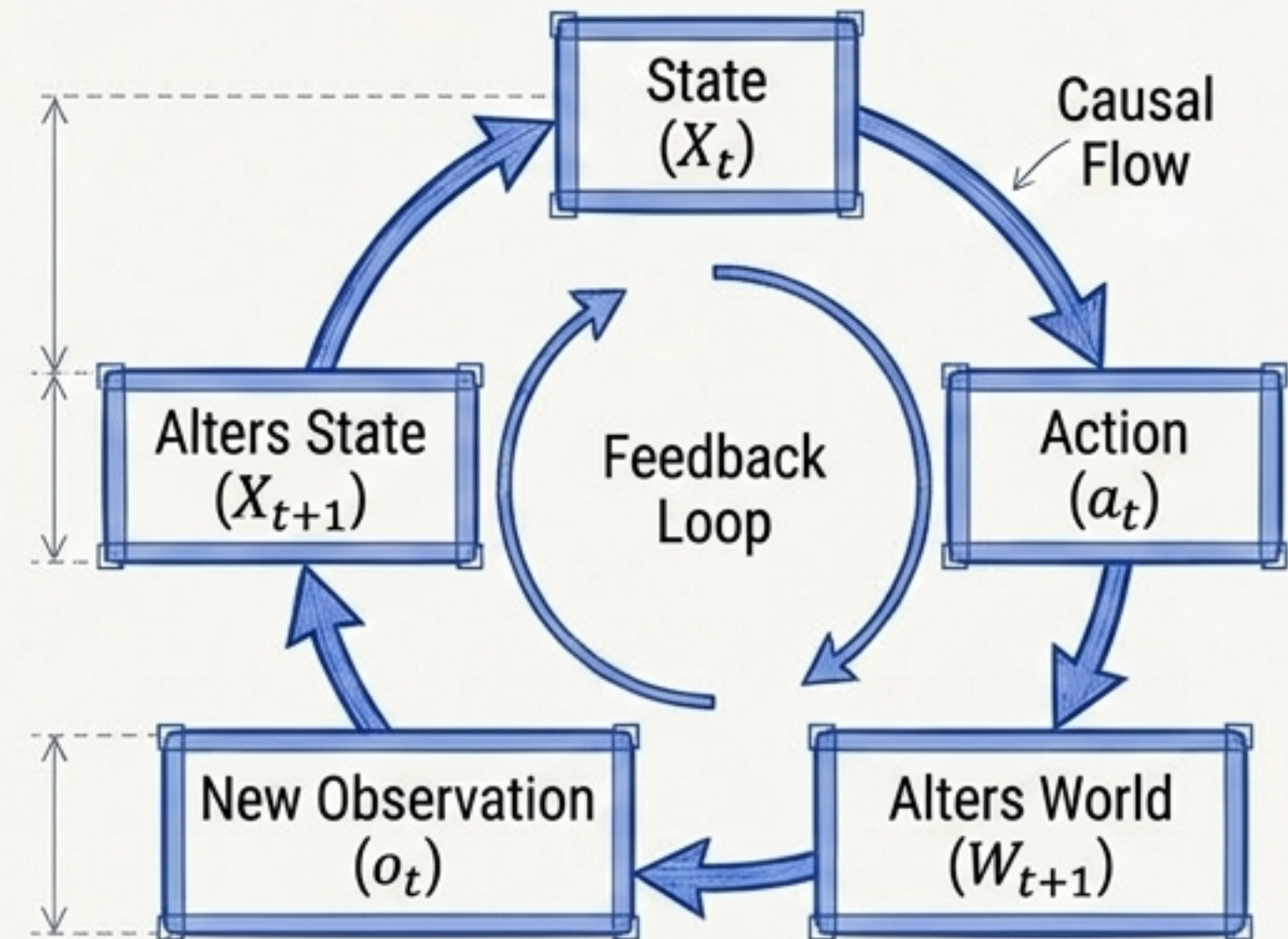
Character Affect: The Closed Loop

A smile does not just reveal happiness; expression is a causal intervention in the world that alters the social environment and future internal states. Character state is therefore non-identifiable from a single frame and inherently history-dependent.

The Pneumatic Model

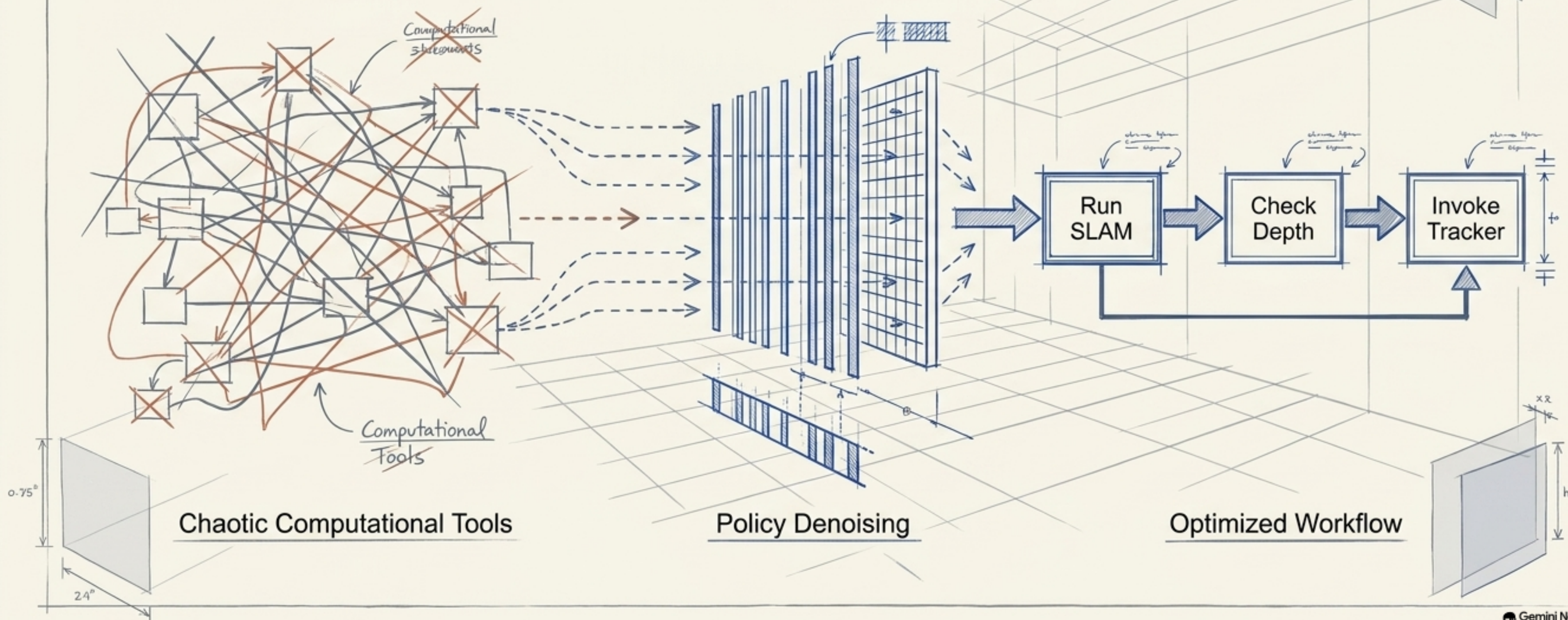


The Closed-Loop Model



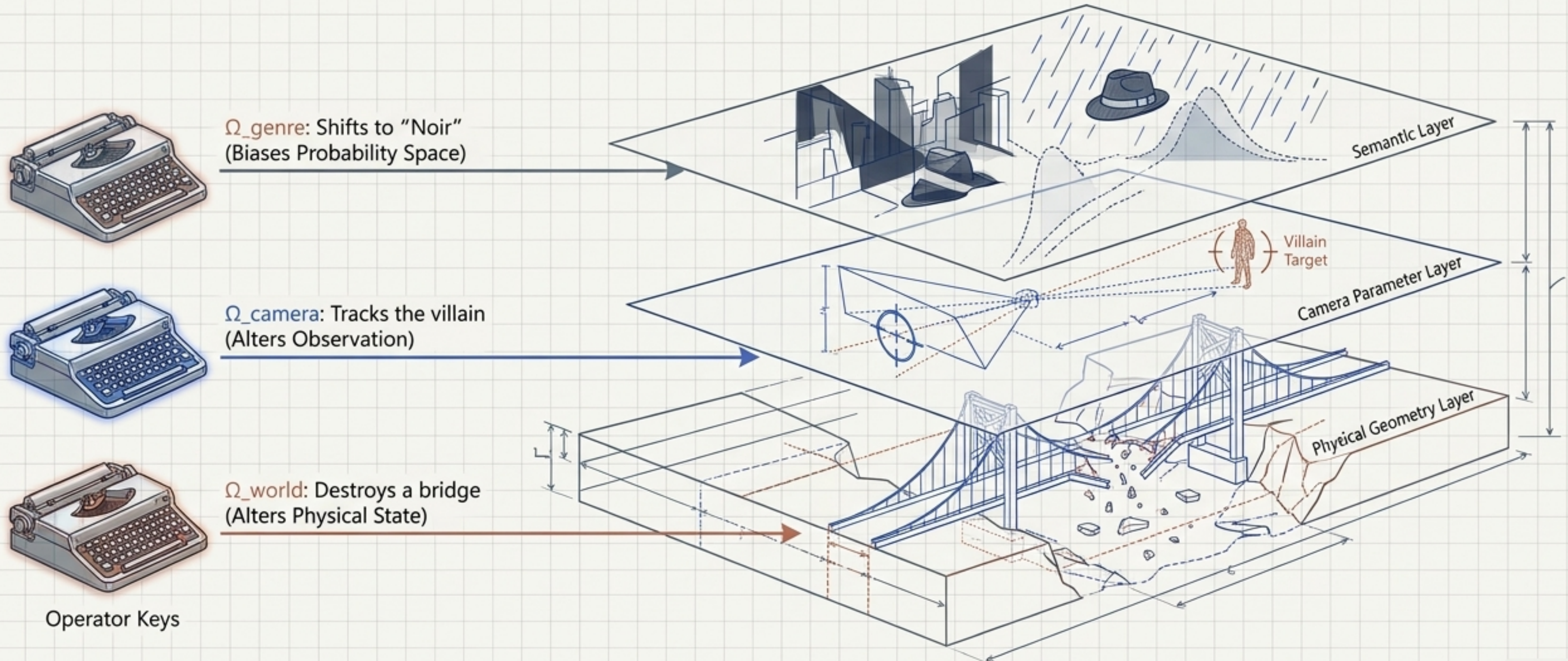
Latent Diffusion Over Policy Space

Generative AI doesn't just have to generate pixels. In this framework, Diffusion is a stochastic proposal mechanism for computational strategy. Models analyze the world state and generate the optimal workflow—which algorithms to run, in what order—to construct the next admissible scene.



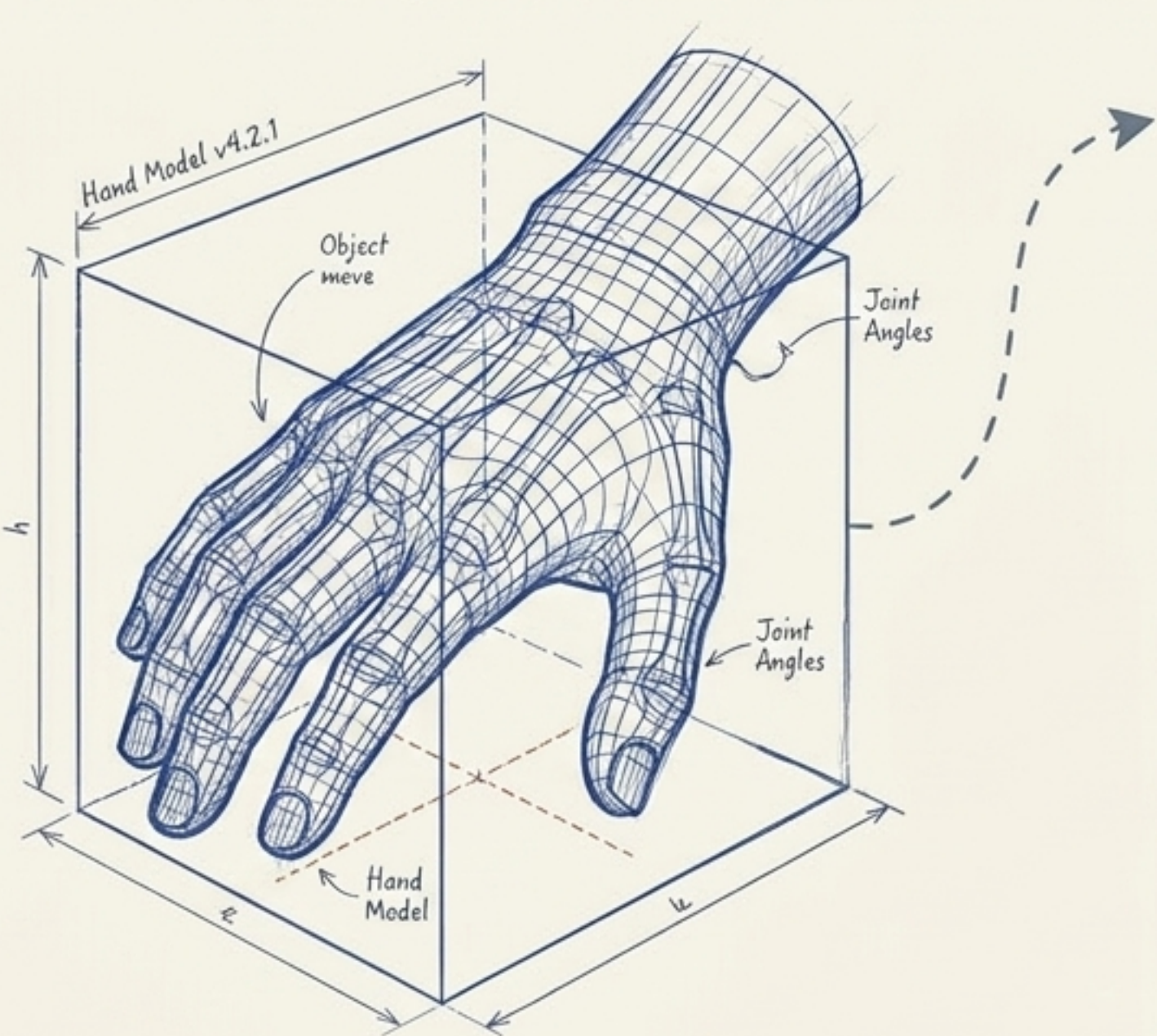
Operator Algebra for Interactive Cinema

Interactive cinema is not a “choose your own adventure” menu. It is the continuous application of mathematical operators modifying narrative grammar, style realizations, and world physics. Because operators are non-commutative (Rain then Noir $\neq$ Noir then Rain), the sequence of user input carries deep semantic meaning.



Evidentiary History & Provenance (H_t)

A system capable of immense generation requires immense accountability. The World State maintains an append-only evidentiary ledger. We don't just know what the scene looks like; we have cryptographic-style proof of why the system believed it was structurally sound.



Timestamp	10:42:01.05
Proposer Process	Temporal Diffusion Model
Witnesses Consulted	Optical Flow, Depth Estimator
Dispersion Metric (Γ)	0.04 (Consensus reached)
Constraints Checked	Rigid Body Interpenetration, Object Permanence
Final Outcome	COMMITTED

The Frame is Dead. Long Live the World.

Cinematic Reconstruction unites three seemingly distinct domains. By elevating our architecture from 2D frames to a persistent, causal 3D reality, we unlock generation without hallucination, diagnosis without ambiguity, and compression without boundary loss.

